



An improved plagiarism detection scheme based on semantic role labeling

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ABSTRACT

Plagiarism occurs when the content is copied without permission or citation. One of the contributing factors is that many text documents on the internet are easily copied and accessed. This paper introduces a plagiarism detection technique based on the Semantic Role Labeling (SRL). The technique analyses and compares text based on the semantic allocation for each term inside the sentence. SRL is superior in generating arguments for each sentence semantically. Weighting for each argument generated by SRL to study its behaviour is also introduced in this paper. It was found that not all arguments affect the plagiarism detection process. In addition, experimental results on PAN-PC-09 data sets showed that our method significantly outperforms the modern methods for plagiarism detection in terms of Recall, Precision and *F*-measure.

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1. Introduction

Plagiarism defined as “unacknowledged copying of documents or programs” Joy [1]. It can occur in many sectors. For example, companies may look for competitive advantage, and academicians need to advance their institutions by searching for quick ways for publishing their works. Most empirical studies and analysis were undertaken by the academic community to deal with student plagiarism. In order to discriminate plagiarized documents from non-plagiarized documents, a correct selection of text features is a key aspect. There are many types of plagiarism mentioned by Hermann Maurer et al. [2], such as copy and paste, redrafting or paraphrasing of the text, plagiarism of idea, and plagiarism through translation from one language to another.

Nowadays, many documents are available on the internet and are easy to access. Due to this availability, users can easily create a new document by copying and pasting from these resources. Sometimes users can reword the plagiarized part by replacing the word with their synonyms. This kind of plagiarism is difficult to be detected by the traditional plagiarism detection system such as COPS [3], SCAM [4], CHECK [5], etc. Owing

to the amount of plagiarized matter, an organization has been built to avoid dishonest works. According to literature review by BAO [6], the current plagiarism detection system was found to be too slow and takes too much time for checking. The matching algorithms are also dependent on the text's lexical structure rather than semantic structure. Therefore, it becomes difficult to detect the text paraphrased semantically. The big challenge is to provide plagiarism checking with appropriate algorithm in order to improve the percentage of finding result and time checking. The important question for the plagiarism detection problem in this study is whether it is possible to apply new techniques such as Semantic Role Labeling to handle plagiarism problems for text documents.

Semantic Role Labeling (SRL) is one of the Natural Language Processing techniques that are used in many fields such as text summarization [7], text clustering [8] and text categorization [9]. In this paper, we proposed a new plagiarism detection method based on SRL and improved similarity measure using argument weighting. The proposed method can detect copy paste plagiarism, rewording or synonym replacement, changing of word structure in the sentences, modifying the sentence from passive voice to active voice and vice versa. SRL was used to analyze the sentences semantically and WordNet thesaurus was used to extract the concepts or synonymies for each word inside the sentences. Weight score is calculated for each argument to study their behaviour and

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effect in plagiarism detection. The rest of the paper is organized as follows: Section 2 provides a description of the related work in plagiarism detection. In Section 3, a full description of the underlying idea involved in our method is covered. Section 4 discusses the experimental design used in our proposed method. Corpus and dataset, including similarity detection and results discussion of the proposed approach, are presented in Section 5, whereas Section 6 concludes the paper.

2. Related works

This section mainly discusses some of recently proposed plagiarism detection techniques. These techniques can be classified into character-based methods, structural-based methods, cluster-based methods, syntax-based methods, cross language-based methods and semantic-based methods.

2.1. Character-based methods

2.1.1. Fingerprint

According to Alzahrani Survey on plagiarism detection methods [10], common plagiarism detection techniques rely on character-based method to compare the suspected document with the original document. Identical string can be detected exactly or partially using character matching approach. Lyon et al. [11] considers text comparison based on word n-grams. With reference to this, the suspected text is split into two sets of tri-grams to be compared. The amount of common tri-grams is considered in order to detect potential plagiarism cases. Kang et al. [12] considers the sentence as the comparison unit in order to compare local similarities. It differentiates among exact copy of sentences, word insertion, word removal and rewording. Heintze [13], Broder [14], and Kriszti [15] proposed a fingerprint method to find the string matching and plagiarism detection based on common fingerprints proportion. These methods obtain good results but fail when the plagiarized part is modified by rewording or changing some words.

2.1.2. String similarity

Brin et al. [3] introduced a plagiarism detection system from Stanford Digital Library Project named COPS (copy protection system), which detects document overlap relying on string matching and sentences. Its main drawback is that it fails to consider individual words and takes the whole sentence as one part. The shortcomings of COPS was solved by Shivakumar and Garcia-Molina [4], who then developed a new method called Stanford Copy Analysis Method (SCAM) to improve the COPS using Relative Frequency Model (RFM) to stand out subset copies. RFM is an essential asymmetric similarity measure for plagiarism detection. The main advantage of SCAM is that it can find the overlapping similarity between the parts of sentences, but many terms can be misleading in documents sharing comparison. The method proposed by Hunt and Szymanski [16] and adopted by Chow and Salim [17] is called the Longest Common Subsequence (LCS). LCS is one of the techniques used in ROUGE, which is a well-known summary evaluation method. Given two sequences X and Y, the longest common subsequence (LCS) for X and Y is the common subsequence with maximum length [18]. White and Joy [19] proposed a new algorithm for plagiarism detection. It is capable of detecting sophisticated obfuscation (such as paraphrasing, reordering, merging, and splitting sentences) as well as direct copying.

2.2. Structural-based methods

It is worth noting that all the studies above described character-based methods. In fact, all these methods focused on lexical features of the text in the document. Many studies that proposed different

methods in the field of plagiarism detection focused on structure features of the text in the document such as headers, sections, paragraphs, and references. Tree-structured features representation is one of the recent studies focused on structure features. It is a rich multi-layer self-organizing maps model (ML-SOM) for text documents [20,21]. Chow and Rahman [22] adopted a tree-structured representation with ML-SOM for information retrieval and plagiarism detection. They built their idea based on two stages, top layer and bottom layer. The top layer presents clustering and retrieving of documents while the bottom layer utilizes a cosine similarity coefficient to capture similar and plagiarized text.

2.3. Cluster-based methods

Document clustering technique is one of the information retrieval techniques that are used in many fields such as text summarization [23], text classification [24] and plagiarism detection [25]. It is used to improve the retrieval data using reduction of the searching time in document location for text summarization and reducing the comparison time in plagiarism detection. Another approach by Si et al. [5] and Zini [26] uses specific words (or keywords) to find similar clusters between documents. Zini [26] also mentioned that the fingerprinting techniques mostly rely on the use of K-grams. Since the process of fingerprinting divides the document into grams of K-lengths, the fingerprints of the two documents can be compared in order to detect plagiarism. Fingerprint-based technique was used for document clustering to summarize a collection of documents and build a fingerprint model for it.

2.4. Syntax-based methods

Elhadi and Al-Tobi [27] introduced a duplicate detection technique for syntactical structures of documents. This technique looks at using syntactical part-of-speech (POS) tags to represent a text structure as a basis for further comparison and analysis. This technique ordered and ranked the documents using POS tags. Elhadi and Al-Tobi [28] improved the methodology of Elhadi and Al-Tobi [27] to calculate the similarity between the documents and ranked them according to the most relevant extracted documents.

Some studies such as Koroutchev and Cebrian [29] compressed the syntax of two texts based on a normalized Lempel–Ziv (LZ) distance algorithm and calculated the similarity for shared topological information given by the compressor. The technique was able to detect similar text documents even if they have different literals. In addition, this method and other methods such as Elhadi et al. [27,28] worked on text reduction using tokenization and stop words removal, and are only interested in smaller sets of syntactical tags.

2.5. Cross language-based methods

A cross-lingual method for detecting suspected documents plagiarized from the language was proposed by Grozea and Popescu [30]. The similarity among suspected and original documents is evaluated using a statistical model. It establishes the probability that a suspected document is relevant to the original document regardless of the order in which the terms appear in the suspected and original documents. Their method is combined with a dictionary corpus of text in English and Spanish to detect similarity in cross language. This approach requires the construction of the cross-lingual corpus, which is a difficult task as mentioned in Grozea and Popescu [30].

2.6. Semantic-based methods

Many researchers have done excellent work to calculate the semantic similarity between words and relatedness between

documents in general using WordNet. Knowledge-based measures [31] identify the semantic similarity between two words by calculating the degree of relatedness among words using information from a dictionary or thesaurus. It makes use of the relations and the hierarchy of a thesaurus, which is generally a hand-crafted lexical database such as WordNet. In our knowledge [32] text documents can be represented as a graph. Then plagiarism can be detected through the graph matching between the concepts of words. A semantic plagiarism detection technique introduced by Alzahrani and Salim [33] uses fuzzy semantic-based string similarity. The method was developed via four main steps. The first step is pre-processing which comprises tokenization, stop words removal and stemming. The second step is retrieval of a list of candidate documents for each suspicious document using *Jaccard coefficient* and shingling algorithm. Suspicious documents are then compared sentence-wise with the associated candidate documents. This step entails the computation of a degree of fuzzy similarity that ranges between two edges: 0 for completely different sentences and 1 for identical sentences. Two sentences are marked as similar if they gain a fuzzy similarity score above a certain threshold. The last stage is post-processing in which consecutive sentences are joined to form single paragraphs or sections.

A different method of semantic-based plagiarism detection was proposed by Chow and Salim [34]. The proposed method calculates the similarity between the suspected and original documents according to the predicates of the sentences. Each sentence predicate is extracted using the Stanford Parser Tree (SPT). The degree of similarity between the extracted predicates is calculated using the WordNet thesaurus. The drawback of this method is that it does not cover all parts of the sentence, only subject, verb and object.

We show through the literature review in this section that many efforts have been made in the past to detect the similarity between text documents. But these methods still need to improve the detecting technique to capture plagiarized parts, especially in semantically plagiarized parts.

The main difference between our proposed method and other techniques is represented in three points: first, our proposed method is a comprehensive plagiarism detection technique, which focuses on many types of detection such as detecting copy paste plagiarism, rewording or synonym replacement, changing of word structure in the sentences and modifying the sentence from passive voice to active voice and vice versa. The second point is corresponds with the comparison mechanism, where our proposed method is taken from the SRL as a means of analysis and comparison to capture a plagiarized meaning of text where the other methods focused on traditional comparison techniques such as character-based and string matching. The important point of this difference is an improvement of similarity score by our proposed method using the SRL weighting scheme where none of the proposed methods are used before.

3. Semantic Role Labeling (SRL)

Semantic structures or semantic frames were introduced by Fillmore [35] where common frames were used for common roles and themes such as FrameNet proposed by Baker et al. [36] and PropBank proposed by Palmer et al. [37]. A statistical system is trained on the data from the FrameNet project to automatically assign semantic roles [38]. Surdeanu et al. [39], Pradhan et al. [40] and, Xue and Palmer [41] followed this approach by improving sets of features and machine learning methods. Barnickel et al. [42] introduced a large scale application of neural network based on Semantic Role Labeling for automated relation extraction from biomedical texts. This method mainly used SENNA software that is used in some application on text documents. SENNA can extract the arguments of the sentences and semantic role for the terms

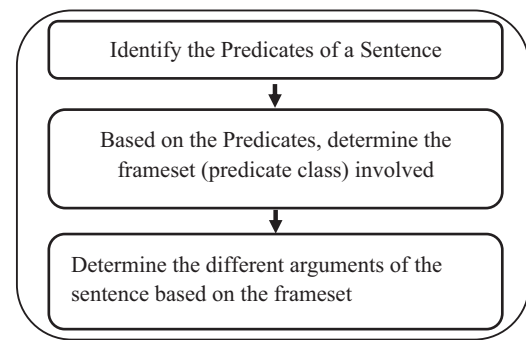


Fig. 1. General process of SRL.

based on neural network algorithms where the authors adopted this software to extract the semantic relations between terms in biomedical text.

SRL is a process to identify and label arguments in a text. The basic idea is that the sentence level semantic analysis of text determines the object and subject of a text. It can be extended to the characterization of events such as determination of “who” did “what” to “whom”, “where”, “when”, and “how”. The predicate of a clause (usually a verb) establishes “what” took place, and other parts of the sentence express the other arguments of the sentence (such as “who” and “when”). The primary task of semantic roles labeling is to identify what semantic relation holds among a predicate and its associate participants or properties, with these relations drawn from a pre-defined list of possible semantic roles for that predicate or class of predicate. The typical labels used in SRL are Agent, Patient and Location for the entities participating in an event. Those labels can be extended to more specific arguments such as time and place in some text. The general process of SRL is shown in Fig. 1 below:

Nowadays, there are several tools developed in assisting Semantic Role Labeling including the Proposition Bank (Propbank), FrameNet [36] and VerbNet [43]. PropBank has thus far obtained the attention of the researchers in SRL technique. The next Section explains the PropBank corpus that is mainly used in SRL.

4. Plagiarism detection using SRL

Plagiarism detection using Semantic Role Labeling aims to detect the semantic similarity between a sentence and possible semantic similarity between two sentences. In this Section, we discuss the idea of our proposed method. We first pre-processed suspected documents and original documents using text segmentation, stop words removal and stemming. Then, SRL was used to transform the sentences into arguments based on location for each term in the sentences. The verbs of the sentences play an important role in the process, and the analysis of the sentences rely on the verbs of the sentences as mentioned in the literature review. Extracted concepts from the arguments were represented as a graph. All the arguments extracted from the text were grouped in the nodes according to the argument type. Each node contains similar extracted arguments. Each group was named by the argument name such as Arg0, Arg1, V, Time, Location, etc. This step is called Argument Label Group (ALG). Then, we extracted all the concepts for each term in the argument groups using WordNet thesaurus. This step is called Semantic Term Annotation (STA). All the concepts that were extracted by WordNet thesaurus were collected in one node known by Topic Signature Node. The advantage of a Topic Signature Node is that it quickly guides us to capture the suspected parts from the documents. Fig. 2 shows general architecture for our proposed method.

Fig. 2 demonstrates the proposed method of general architecture, which includes the following stages.

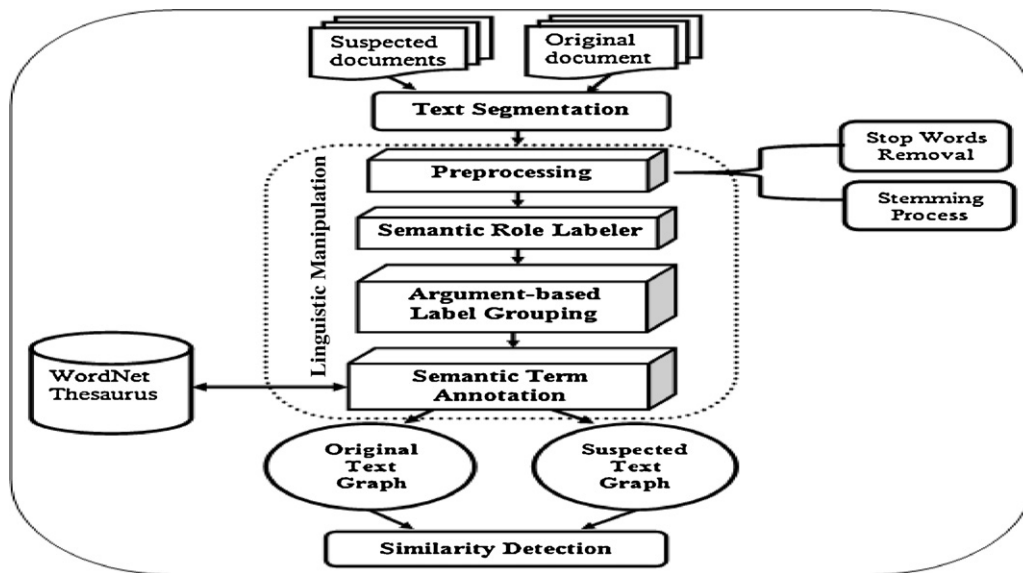


Fig. 2. Proposed method of general architecture.

4.1. Text segmentation

Pre-processing is one of the key steps in Natural Language Processing (NLP). A simple type of pre-processing is text segmentation. The process of dividing text into meaningful units is called text segmentation. Text can be divided by sentences, words or topics. We used sentence-based text segmentation as a first step in our proposed method, where the original and suspected texts are divided into sentence units. This is because our proposed method aims to compare a suspected text with original text based on the sentence matching approach.

4.2. Stop words removal and stemming process

Technology of stop words removal for deleting meaningless words was used. Stemming algorithm was also applied to remove the affixes (prefixes and suffixes) in a word in order to generate its root word. The stop words are the words that frequently occur in documents. These words do not give any hint values or meanings to the content of documents, such as *the*, *and*, and *a*; hence they are eliminated from the set of index terms [44]. Tomasic and McGill [45] reported that such words comprise around 40 to 50% of a collection of documents text words. Eliminating the stop words in automatic indexing speeds the system processing, saves a huge amount of space in index, and does not damage the retrieval effectiveness [46]. There are various approaches used for determination of such stop words. Nowadays, there are several English stop words lists that are commonly used in information retrieval. Our proposed method was carried out to eliminate all the stop words in the text to speed up the system processing.

On the other hand, another text pre-processing is stemming. Stemming is a solution for one of the problems involved in information retrieval, like variation in word forms [47]. The most common types of variation are spelling errors, alternative spelling, multi-word construction, transliteration, affixes and abbreviations. These variations in words form lead to the efficiency issue in the matching algorithm during the information retrieval process. Using root word in pattern matching provides a much better effectiveness in information retrieval. We used this step in our proposed method to extract the significant words from the text and ignore the remaining words. This may adversely affect the similarity between documents in our proposed method.

4.3. Argument-based label grouping

This step represents the sentences in the form of nodes. SRL was used to extract the arguments and role of each term in the argument from the sentences. Extracted concepts from these arguments were represented as a graph. All the arguments extracted from the text will be grouped in the nodes according to the argument type. Each node type contains similar extracted arguments. Each group was named by the argument name such as Arg0, Arg1, V, Time and Location, and this step is called Argument Label Group (ALG).

4.4. Semantic Term Annotation

The concepts that were extracted from each term in the argument groups using WordNet thesaurus is called Semantic Term Annotation (STA). Concept extraction is an important step in our proposed method where different concepts having the same meaning can be referred to as one term which makes the text easier to be checked for plagiarism. In this step, concept extraction was carried out using a WordNet thesaurus by extracting the hypernyms and synonyms for the terms. We believe that by using STA, the difficulty of detecting plagiarized text will be easier, especially for plagiarism using paraphrasing or rewording. Fig. 3 illustrates the phases of our proposed method.

Fig. 3 demonstrates the structure phase of our proposed method deeply. All the parts of the general architecture that were illustrated in Fig. 2 appeared this figure.

The terms in the original topic signature and the suspected topic signature were compared. When we find two terms are identical, we look directly to the argument label group that contain those terms, and then compare the sentences that convey these terms. This step compares the argument labels of possible sentences that have been plagiarized with corresponding argument labels in original sentences. The argument label group guides us to the main argument and each argument inside the group gives a quick guide to the possible plagiarized sentence. The comparison between the terms must be done in the right way. If we compare the terms in Arg0 (subject) in the suspected text with all the other arguments in the original text to determine the plagiarism ratio, it can be wrong. For instance, it is not fair to compare the subject with Time argument (Arg-TMP) with the subject with Adjective argument (Arg-O).

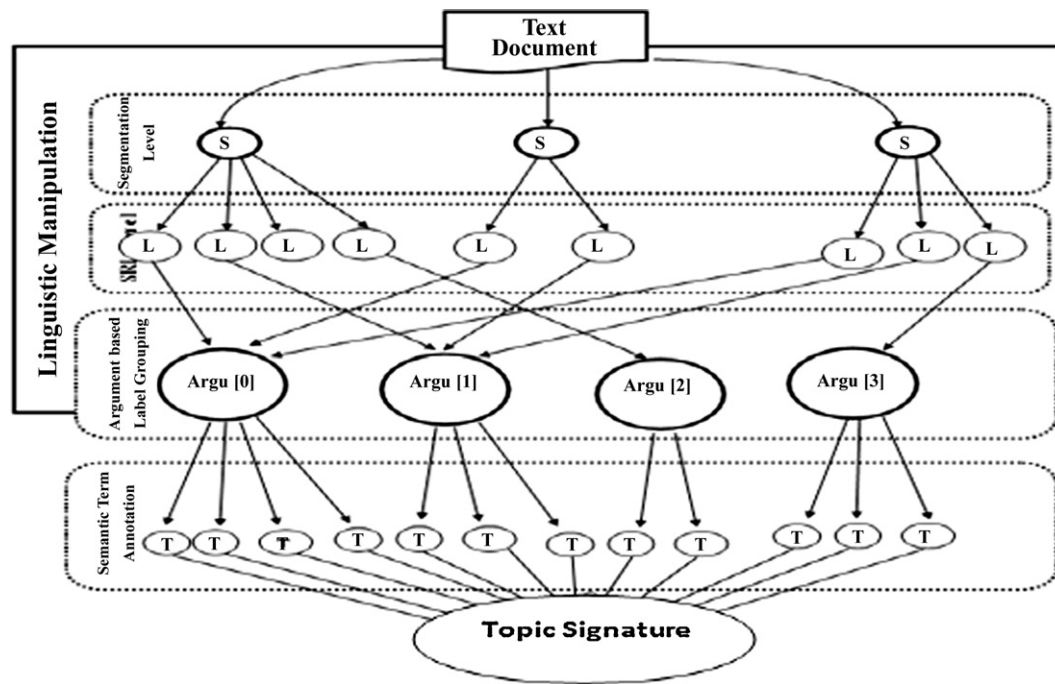


Fig. 3. Structure phase of the proposed method.

The example below explains the right way to compare between the terms.

Suppose we have the following sentences:

John eats food.

Mary drinks water.

John (Arg0)

Eat (verb)

Food (Arg1)

Mary (Arg0)

Drink (verb)

Water (Arg1)

In this example, the comparison used in many techniques such as string matching [12] or n-gram [21] compares each word in the suspected sentence with each word in the original sentence. The term "John" will be compared with the term "Mary", "Drink" and "Water". Not only is this comparison improper, but it also takes time for comparison. The function of our proposed method is to focus on the comparison between the arguments of the suspected sentence with similar arguments in the original sentence. When we group all similar arguments together, we can compare subject with subject, verb with verb, etc. This will lead us to decrease the number of comparisons. Each argument in suspected documents will only be compared with a similar argument in original documents, except Arg0 and Arg1, which will be taken as a special case. Plagiarists can change the place of Arg1 (object) with Arg0 (subject) and vice versa. Fig. 4 shows the similar arguments comparison between suspected and original documents.

Fig. 4 demonstrates the similarity operations between the arguments. We can see in this figure that the comparison operation can be applied between the similar arguments and how the topic signature plays an important role in similarity calculation.

Another situation for the plagiarist can be shown through the following example:

John gave Mary the book (original sentence).

The book was given to Mary by John (suspected sentence).

By using cognitive computation group SRL tool [51], the produced arguments are:

John gave Mary the book

The book was given to Mary by John

Figs. 5 and 6 illustrate the analysis for suspected sentence using SRL in the example above. We note that the structure of two sentences above may differ if the active versus passive voice or

synonyms and antonyms are used. Actually, these sentences can be semantically the same. It was noted that the SRL captures the arguments (subject, object, verb and indirect object) for a sentence despite changing the places for the labels inside the sentences. This capturing supports our proposed method in plagiarism detection if comparison is applied based on the arguments of the sentence using SRL.

5. Experimental design and dataset

Our experiments looked at the amount of detected plagiarized sentences from the original documents. The experiments were performed on 100 suspected documents, each plagiarized from one or more original documents according to the PAN-PC-09 dataset. We applied our technique by searching for the suspected documents within original documents. The documents were divided into five groups. Each group had a certain number of documents. The documents increased for each group with each testing of comparison. We started by 5 documents in the first group. Then, we added 5 more documents to the first group and 10, 20, 40, and 100, respectively. The aim of this grouping process was to study the behaviour of the plagiarized user for each argument so it can be focused. As an initial step, we ran some experiments on the arguments that were extracted from SRL and we noted that some of the arguments conveyed a very small value or had inverse proportion and some of them had a direct proportion between the number of documents, and the similarity score when we calculated the similarity. These indicators reflected the decreasing and increasing similarity score between the suspected documents and original documents. The important arguments were then selected to enter the second testing comparison process. On the other hand, we ignored the arguments that had an inverse proportion. Testing was applied another time after arguments selection. We found that the score of similarity was declining as compared with the first testing because the degree of similarity depended on the number of arguments extracted from the sentences, and the reducing of the unimportant arguments consequently led to the rise of the similarity score and vice versa. The similarity score

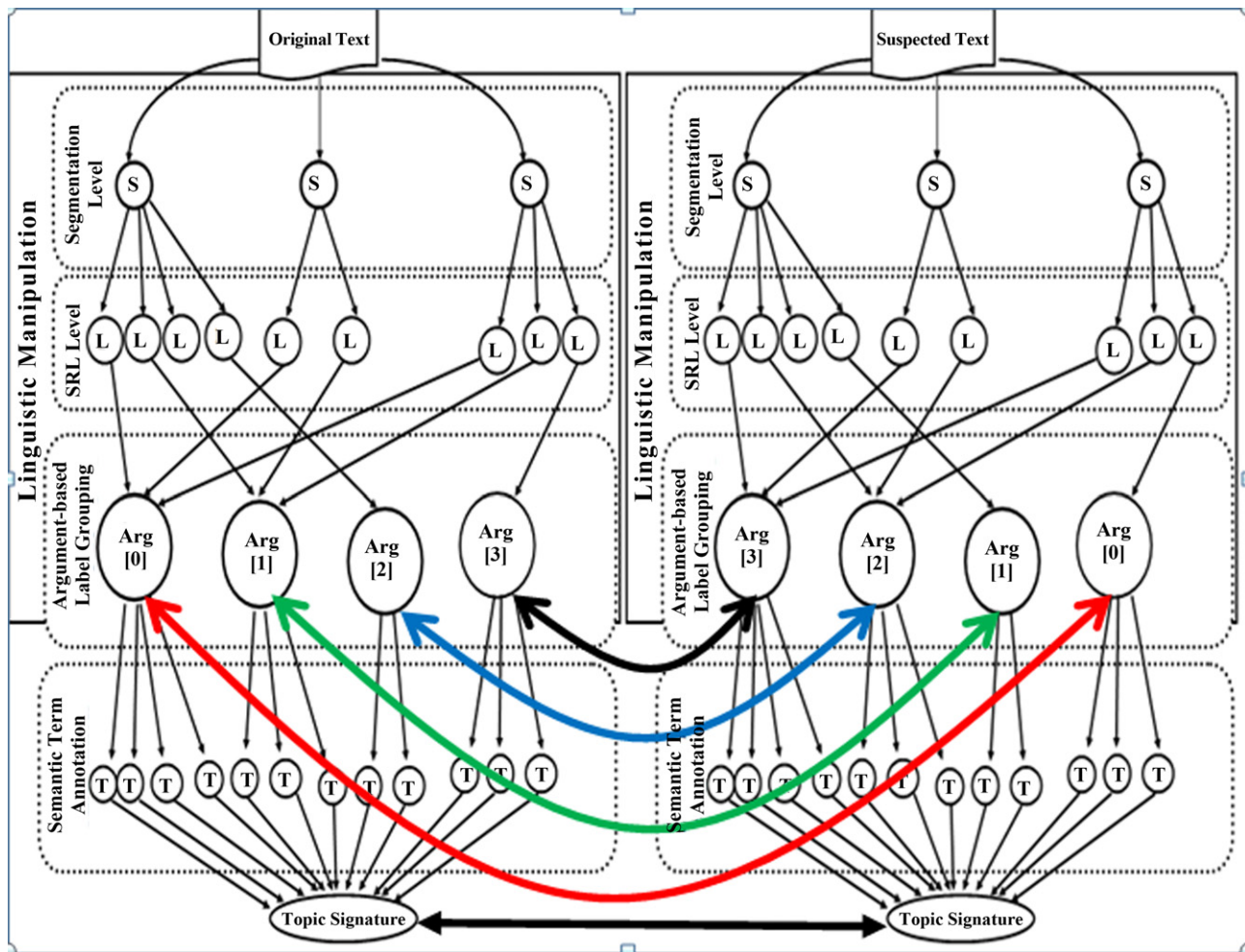


Fig. 4. Similar arguments comparison.

was calculated, and then the PAN-PC-09 plagiarism dataset was used for cross-checking. The details of similarity calculation and arguments weight will be explained in the next sections.

5.1. Similarity detection

In this stage, sentence-based similarity analyses between tokenized suspected and original documents were performed. Sentences in suspected documents were compared with each sentence in the candidate documents according to the arguments of the sentences. We aimed to detect not only the arrangement similarity between sentences, but also possible semantic similarity between two sentences.

Similarity detection was conducted by comparing the original Topic Signature terms and suspected Topic Signature terms. If the

two terms were found to be identical, we went directly to the argument label groups that contained these terms, and then determined the label group where they belonged, thus determining the possible sentences that may be plagiarized. This step compared the arguments of possible sentences that had been plagiarized with the corresponding arguments in original sentences. The argument label group guided us to the main arguments and each argument inside the group quickly guided us to the possible plagiarized sentence.

Some variables play an important role in the similarity calculation, such as the number of matched arguments and number of arguments which exist in the sentences. The first variable determines the similar arguments between the suspected document and original document while the second variable determines the

Output:

	SRL	Charniak
John	giver [A0]	(S1 (S (NP (NMP John)))
gave	V: give	(VP (VBD gave)
Mary	entity given to [A2]	(NP (NMP Mary))
the	thing	(NP (DT the)
book	given [A1]	(NN book)))
.		(. . .))

Fig. 5. Analysis for original sentence using SRL.

Output:

	SRL	Charniak
The	thing	(S1 (S (NP (DT The)
book	given [A1]	(NN book)))
was		(VP (AUX was)
given	V: give	(VP (VEN given)
to	entity	(PP (TO to)
Mary	given to [A2]	(NP (NMP Mary)))
by	giver [A0]	(PP (IN by)
John		(NP (NMP John))))
.		(. . .))

Fig. 6. Analysis for suspected sentence using SRL.

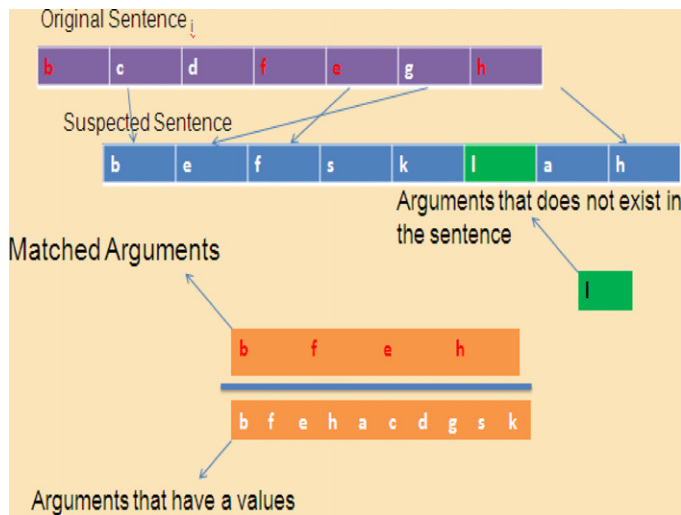


Fig. 7. Similarity calculations between the suspected and original document.

argument that does not exist in the sentences. The similarity between the arguments of the suspected document and original document was calculated according to *Jaccard coefficient* that can be defined in the following equation:

$$\text{Similarity } C_i(\text{ArgS}_j, \text{ArgS}_k) = \frac{C(\text{ArgS}_j) \cap C(\text{ArgS}_k)}{C(\text{ArgS}_j) \cup C(\text{ArgS}_k)} \quad (1)$$

where, $C(\text{ArgS}_j)$ = concepts of the argument sentence in the suspected document; $C_i(\text{ArgS}_k)$ = concepts of the argument sentence in the original document;

We then calculated the similarity between the suspected document and original document based on the following equation:

$$\text{Total similarity } (\text{Doc1}, \text{Doc2}) = \sum_{i=1, l} \sum_{j=1, m} \text{SimC}_i(\text{ArgS}_j, \text{ArgS}_k) \quad (2)$$

where, $\text{SimC}_i(\text{ArgS}_j, \text{ArgS}_k)$ is similarity between Arguments sentence j in suspected document containing concept i Arguments sentence k in original document containing concept i , l = no. of concepts, m = no. of Arguments sentence in suspected document, n = no. of Arguments sentence in original document.

Fig. 7 below shows the similarity calculation between the suspected document and original document according to their arguments.

Fig. 7 demonstrates the similarity calculations between the suspected and original document. As mentioned in Section 4, similarity was calculated between all corresponding arguments. We noted in Fig. 7, arguments b, f, e, and h were overlapping between the original and suspected sentences where these arguments are called Matched Arguments. We also observed in the same figure that we have an argument that does not exist in the original sentence but existed in suspected sentences such as Argument l. Actually, this Argument is dissimilar for original arguments in order to decrease

Table 2
Argument types and their descriptions.

Type	Description
Arg0	Agent
Arg1	Direct object/theme/patient
Arg2–5	Not fixed
V	Verb
MNR	Manner
TMP	Time
DIS	Discourse connectives
ADV	General purpose
NEG	Negation marker
LOC	Location
PNC	Purpose
MOD	Modal verb
O	Adjective
DIR	Direction
EXT	Extent

a similarity score between the original and suspected sentences and thus the similarity score is equal to zero. In this case, our proposed method ignored these types of argument in similarity calculation and only arguments that had value were taken in the similarity calculation.

5.2. Results and discussion

Our technique was tested according to the group of documents (5, 10, 20, 40, and 100). Those suspected documents were plagiarized with different ways of plagiarism such as simple copy and paste, changing some terms with their corresponding synonyms, and modifying the structure of the sentences (paraphrasing). We provided three general testing parameters that are commonly used in plagiarism detection as following:

$$\text{Recall} = \frac{\text{number of detected arguments}}{\text{total number of arguments}} \quad (3)$$

$$\text{Precision} = \frac{\text{number of plagiarized arguments}}{\text{number of detected arguments}} \quad (4)$$

$$F\text{-measure} = \frac{2 \times \text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} \quad (5)$$

Table 1 illustrates the results across the set of documents. We selected one row for each group to explain the situation of the arguments during the similarity calculation.

Table 1 illustrates the results across the set of documents. We have 19 arguments that have been extracted using SRL. According to the similarity score between these arguments, we observed that many of the arguments have zero score such as NEG (Negative word in sentence such as “not”). This score indicates that the argument does not exist and has zero value after similarity calculation. There are many types of arguments identified by SRL.

Table 2 illustrates these types that appeared in Table 1.

Table 2 demonstrates the types of arguments that were used in our experiments and appeared in Table 1 above. In addition, the description meaning for the arguments is clarified in this Table.

The results of similarity calculation in term of Recall, Precision and *F-measure* are set out in Table 3 below.

Table 1
Results across the set of documents.

Number of documents in set	A0	A0A1	A1	A1A0	A2	Verb	MNR	TMP	DIS	ADV	NEG	LOC	PNC	MOD	O	A3	A4	DIR	EXT
5	0.93	0.74	0.8	0.51	0.82	0.8	0.52	0.75	0	0	0	0.83	0	0	0	0	0	0	0
10	0.82	0.65	0.92	0.52	0.94	0.86	0	0.79	0	0.85	0	0.88	0	0	0.91	0.8	0	0	0
20	0.89	0.67	0.67	0.84	0.77	0.74	0.66	0.85	0.7	0	0	0.66	0.9	0	0.87	0.8	0	0	0
40	0.97	0.78	1	0.85	0.82	0.79	0.75	0.81	0	0.77	0	0	0	0	0.88	0	0	0	0
100	0.94	0.84	0.85	0.87	0.76	0.9	0	0.81	0	0.66	0	0.61	0	0	0.77	0	0	0	0

Table 3
Results after similarity calculation.

Number of documents in set	Recall	Precision	F-measure
5	0.863158	0.586471	0.698409
10	0.826316	0.734412	0.777658
20	0.802632	0.61239	0.694722
40	0.807895	0.678677	0.73767
100	0.818421	0.642406	0.719809

Table 3 demonstrates the results of similarity between the suspected and original documents for each set of documents. We found that all the values in recall measure were above 0.80 and all the values in precision and *F*-measure were more than 0.58. Actually this values seemed a good results because it is greater than 0.50 but we tried to improve these scores to obtain high similarity score.

We calculated the weighting for each argument to optimize the obtained results that were achieved in Table 3 according to the following equation:

$$W Arg_i = \frac{\text{number of Args that have value in group } i}{\text{total number of Docs in group } i} \quad (6)$$

where, $W Arg_i$ = weight of argument in document i . Args that have value in group i = summation of each value of argument in group i ; Total number of Docs in group i = number of documents in group i ; i = (5, 10, 20, 40, 100).

After the optimization process using argument weight, we observed the influence of the weight enforcing our observations in direct and inverse proportion. This observation reflects that the plagiarizing user does not focus on all arguments of the sentences, thus we tend to ignore some arguments. We consider these to be important arguments.

Table 4 demonstrates the behaviours of the arguments after the optimization process. Two types of arguments are depicted in Table 4. The first type of argument has a decreasing similarity score, such as DIS, NEG, PNC, MOD, A3, A4, and EXT while the rest of arguments are a second type. We consider these to be important arguments. We selected these arguments to improve the similarity score by the weighting optimization process. On the other hand, we ignored the first type of argument because the similarity score is going to zero or less, because this type of argument reduces the similarity score in order to reduce the overall similarity ratio between the suspected and original documents. Fig. 8 below shows the behaviour of the arguments after selecting the important arguments in the optimization process.

Table 4
Behaviours of the arguments after the optimization process.

Number of documents in set	A0	A0A1	A1	A1A0	A2	Verb	MNR	TMP	DIS	ADV	NEG	LOC	PNC	MOD	O	A3	A4	DIR	EXT
5	0.72	0.59	0.84	0.47	0.69	0.85	0.65	0.6	0	0.4	0	0.73	0.4	0	0.77	0	0	0.09	0
10	0.67	0.59	0.84	0.53	0.65	0.84	0.5	0.7	0.1	0.5	0	0.63	0.3	0.12	0.8	0.2	0.1	0.24	0.08
20	0.7	0.64	0.79	0.54	0.57	0.72	0.44	0.5	0.1	0.4	0	0.53	0.3	0.11	0.73	0.2	0	0.15	0.04
40	0.76	0.64	0.82	0.6	0.63	0.77	0.41	0.6	0.1	0.4	0	0.65	0.2	0.08	0.81	0.2	0	0.08	0.02
100	0.82	0.72	0.77	0.72	0.66	0.77	0.49	0.7	0.1	0.5	0	0.47	0.2	0.04	0.8	0.1	0	0.07	0.01

Table 5
Results after the optimization process and the selection of important arguments.

Number of documents in set	A0	A0A1	A1	A1A0	A2	Verb	MNR	TMP	ADV	LOC	O	DIR
5	0.72	0.59	0.84	0.47	0.69	0.85	0.65	0.6	0.4	0.73	0.77	0.09
10	0.67	0.59	0.84	0.53	0.65	0.84	0.5	0.7	0.5	0.63	0.8	0.24
20	0.7	0.64	0.79	0.54	0.57	0.72	0.44	0.5	0.4	0.53	0.73	0.15
40	0.76	0.64	0.82	0.6	0.63	0.77	0.41	0.6	0.4	0.65	0.81	0.08
100	0.82	0.72	0.77	0.72	0.66	0.77	0.49	0.7	0.5	0.47	0.8	0.07

Table 6
Evaluation results after selection of the important arguments.

Number of documents in set	Recall	Precision	F-measure
5	0.836364	0.802828	0.819253
10	0.863636	0.79202	0.826279
20	0.777272	0.764477	0.770821
40	0.816364	0.799897	0.808047
100	0.83	0.895167	0.861353

In Fig. 8, the behaviour of the arguments under different number of documents is fluctuating. Some of the arguments increase to the high score 1 and some of them decrease to zero score. The increasing and decreasing of these scores demonstrates the important and unimportant arguments.

Table 5 shows that all the arguments demonstrate a score higher than zero. This indicates that the evaluation score is increasing.

Table 6 demonstrates the evaluation score after the weightings of arguments were optimized. It shows that weightings of arguments give results for plagiarism detection.

We ran some statistical significance tests (*t*-tests) to show the improvements of our proposed method. We found that our proposed method was statistically significant, and the testing values are shown in Table 7 below.

This table displays the number of cases, mean value, standard deviation, standard error and significance values for the pairs of variables before and after optimization using SRL weighting scheme ((Recall1, Recall2); (Precision1, Precision2); and (*F*-measure1, *F*-measure2)) compared in the Paired Samples *T*-Test procedure. The Paired-Samples *T*-Test procedure compares the means of two variables that represent the same group at different times. The mean values for the two variables ((Recall1, Recall2); (Precision1, Precision2); and (*F*-measure1, *F*-measure2)) are displayed in the Paired Samples Statistics Table. Since the Paired Samples *T*-Test compares the means for the two variables, it is useful to know what the mean values are. A low significance value for the *T*-test (typically less than 0.05) indicates that there is a significant difference between the two variables. We noted that in Table 7, in terms of precision (0.010) and *F*-measure (0.006), this condition was emphasized and it failed in Recall (0.936), which means our proposed method obtained significant results in Precision and *F*-measure. The confidence interval for the mean difference does not contain zero; this also indicates that the difference is significant. In addition, the significance value is low in Precision and Recall values and the confidence interval for the mean difference does not

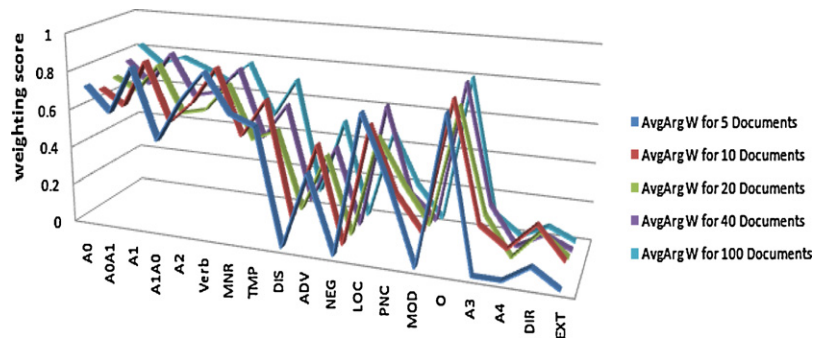


Fig. 8. Behaviours of the Arguments after the optimization process.

Table 7

Statistical significant testing using *t*-test.

		Paired differences					
		Mean	Std. deviation	Std. error mean	95% confidence interval of the difference		Sig. (2-tailed)
					Lower	Upper	
Pair 1	Recall1–Recall2	–1.042800000	2.71762326	1.21535807	–3.478654966	3.27009497	.936
Pair 1	Precision1–Precision2	160006600	7.71870131	3.45190817	–.255846935	–6.416626466	.010
Pair 1	<i>F</i> -measure1– <i>F</i> -measure2	–9.149700000	3.83648693	1.71572912	–.139133277	–4.386072294	.006

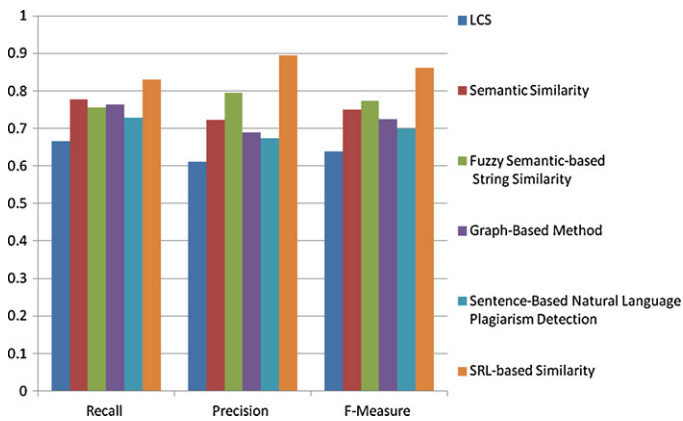


Fig. 9. Comparison results with plagiarism detection techniques.

contain zero. We can therefore conclude that there is a significant difference between results before and after optimization.

Our proposed method is evaluated and compared with some plagiarism detection algorithms in PAN 2010 competition.

Fig. 9 illustrates the comparison between our proposed method and other techniques.

Fig. 9 demonstrates the comparison between SRL-based similarity with fuzzy semantic-based string similarity [33], longest common subsequence (LCS) [48], semantic-based similarity [34], graph-based method [32] and sentence-based natural language plagiarism detection [19]. We found that our proposed method achieved better results in terms of Recall, Precision and *F*-measure.

Another evaluation criterion is time efficiency or time complexity. This measure is commonly used to evaluate the algorithms. We evaluated our proposed method based on time complexity. We found that our proposed method belongs to the $O(n^2)$ class. Many of the plagiarism detection algorithms belong to the $O(n^2)$ class according to studies by Mozgovoy et al. [49] and JPlag [50]. However, they found some of the plagiarism detection systems have $O(f(n)N^2)$, where N is the size (number of files) of the collection, and $f(n)$ is the time to make the comparison between one pair of files of length n . We also compared the time complexity of our proposed method with fuzzy semantic-based string similarity, longest

Table 8

Comparison between our proposed method and other techniques using time complexity.

Algorithm	Time complexity
Fuzzy Semantic-based String Similarity	$O(n^2)$
Longest Common Subsequence (LCS)	$O(n^2)$
Semantic-based similarity	$O(n^2)$
SRL-based Similarity (SRL)	$O(n^2)$
Graph-based Method	$O(V+E)$
Sentence-based Natural Language	$O(n^2)$

common subsequence (LCS), semantic-based similarity, graph-based method and sentence-based natural language plagiarism detection. Table 8 below demonstrates the results in term of time complexity.

Table 8 demonstrates the time complexity comparison between SRL-based similarity with fuzzy semantic-based string similarity, longest common subsequence (LCS), semantic-based similarity, graph-based method and sentence-based natural language plagiarism detection. We found that our proposed method belongs to the $O(n^2)$ class.

The improvement that is provided by our proposed scheme is represented on the selection of the important arguments that can affect on plagiarism detection. Another improvement represented in the comparison mechanism where the comparison between two texts should be made by comparing the corresponding arguments only such as (Subject with Subject, Verb with Verb) not similar to the traditional comparison mechanism (Subject with Verb, Subject with Adverb, Subject with Adjective and so on). The major improvement represented in the type of plagiarism detection. It can detect copy paste and semantic plagiarism, rewording or synonym replacement, changing of word structure in the sentences, modifying the sentence from passive voice to active voice and vice versa.

6. Conclusions and future work

Semantic role labeling can be used for plagiarism detection by extracting argument of sentences and comparing the arguments. The effects of arguments are studied using arguments weighting. Through this weighting, we used only the important arguments

in the similarity calculation process. Tests were carried out using PAN-PC-09 standard dataset for plagiarism detection. The proposed methods were found to achieve better performance compared to fuzzy semantic-based string similarity, longest common subsequence (LCS), semantic-based similarity, graph-based method and sentence-based natural language plagiarism detection.

As future work, we plan to investigate role of the terms in SRL using Fuzzy Inference System (FIS) to enhance the similarity score that improved by the argument weighting schema. For instance, we can develop a way to determine all important arguments that have more contribution than the other arguments in a sentence. We also plan to improve our proposed method by using more documents from the PAN-PC dataset.

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